

Homework B2 | ECON 0100 | Fall 2026 | Solutions

Solution guide for the teaching team. Answers and working are in red; everything in black is what students see. Students answer on paper and submit selections on Gradescope, where the homework is completion-graded.

Butterbeer

The supply curve for butter beer can be represented by the following equation:

$$P = 20 + \frac{1}{2}Q$$

Prices are in galleons and quantity is in bottles.

Q1 | Quantity Supplied

Use a graph to plot this supply curve, including the quantity supplied at both 40 galleons and 50 galleons.

- a) Quantity supplied at 40 galleons: 40 bottles
- b) Quantity supplied at 50 galleons: 60 bottles
- c) How much did quantity supplied change as the price dropped from 50 to 40? down by 20 bottles
- d) What is marginal cost at a quantity of 60? 50 galleons

Plot. Vertical intercept, $Q = 0$: $P = 20$. Below 20 galleons quantity supplied is zero, so the curve starts there and rises by 1 galleon for every 2 bottles.

a) $40 = 20 + \frac{1}{2}Q \rightarrow 20 = \frac{1}{2}Q \rightarrow Q = 40$ b) $50 = 20 + \frac{1}{2}Q \rightarrow 30 = \frac{1}{2}Q \rightarrow Q = 60$

c) $\Delta Q_s = 40 - 60 = -20$: quantity supplied fell by 20 bottles as the price fell, the law of supply.

d) Quantity to price. The height of supply at 60 bottles is the marginal cost of the sixtieth bottle: $P = 20 + \frac{1}{2} \cdot 60 = 50$ galleons. It matches b): 50 galleons is the price at which 60 bottles are supplied.

Q2 | Producer Surplus

Then find and label the producer surplus at these prices.

- a) PS at 40 galleons: 400 galleons
- b) PS at 50 galleons: 900 galleons
- c) How much did producer surplus change as the price dropped from 50 to 40? down by 500 galleons

Producer surplus is the triangle above supply and below the price, out to the quantity sold from Q1. Its height is the price minus the 20-galleon intercept and its base is the quantity.

$$P = 40 : \quad PS = h \cdot b \cdot \frac{1}{2} = (40 - 20) \cdot 40 \cdot \frac{1}{2} = 400$$

$$P = 50 : \quad PS' = h \cdot b \cdot \frac{1}{2} = (50 - 20) \cdot 60 \cdot \frac{1}{2} = 900$$

c) $\Delta PS = PS - PS' = 400 - 900 = -500$: producer surplus fell by 500 galleons, the strip between the two price lines. Revenue is the rectangle below the price: $40 \cdot 40 = 1600$ at 40 galleons and $50 \cdot 60 = 3000$ at 50; the rest of each rectangle covers marginal costs.

